

Off-CPU memory

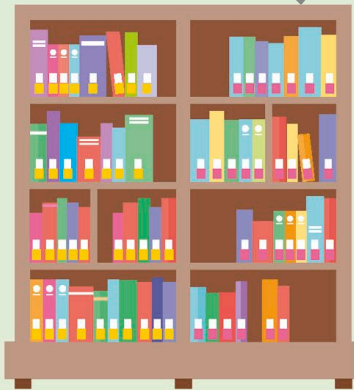
Primary memory includes read-only memory (ROM), containing instructions that start the computer. Most of the primary memory is made up of random-access memory (RAM), which contains data and instructions currently being used. Its contents are lost when the computer is switched off, so long-term data is stored on the hard disk, known as secondary memory.



On-CPU memory ▸

The CPU itself contains registers, where data currently being used is stored, and a cache containing data and instructions likely to be reused soon.

Information is contained in memory, just like books arranged in a library.



Memory

◁ Buses

To transfer information to the CPU, a computer has dedicated electrical connections called buses. The data bus carries data and the address bus carries the addresses of data in memory.

Output



What makes a computer powerful?

Computing power depends primarily on a combination of how fast the CPU can work and how much data it can store in its primary memory. It's also influenced by how fast data can be moved on the computer's buses, and how long it takes to access its secondary memory. If the computer is to be used to process a lot of graphics, its speed can be improved by adding a video card containing a processor optimized for handling images.



△ Benchmarking

Running a set of standard tests on a computer to evaluate how quickly it completes them is called benchmarking. This makes it possible to compare the performance of different processors.

IN DEPTH

Clock-speed

The clock-speed of a processor is a measure of how many instructions it can carry out per second. This is measured in megahertz (MHz) or gigahertz (GHz). A 1.5 GHz processor can carry out 1.5 billion instructions a second. It is possible to override the setting that determines clock-speed on a computer to make it run faster. This is called over-clocking, but it can cause data corruption and damage to the computer through overheating.

